

Tirthajit Baruah

PhD in Cyber Physical Systems, IISc
BS-MS in Electrical Engineering and Computer Science
Minor in Data Science and Engineering, IISER Bhopal
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Education

Indian Institute of Science, Bangalore

PhD, Cyber Physical Systems

2023–Present

Indian Institute of Science Education and Research, Bhopal

Bachelor of Science – Master of Science

2018–2023

CGPA: 8.3/10

Publications

1. **T. Baruah**, K. Jamadar, P. Rathore, “*Two-Fold Patch Perturbation for Efficient Self-Supervised Learning in 3D Medical Imaging.*” Accepted at the International Joint Conference on Artificial Intelligence (IJCAI-ECAI), Bremen, 2026. [LINK]
2. **T. Baruah**, P. Rathore, “*Translating Classifier Scores into Clinical Impact: Calibrated Risk and Queueing Simulation for AI-Assisted Radiology Worklist Triage.*” Accepted at DAI & AIMedHealth @ Association for the Advancement of Artificial Intelligence (AAAI), Singapore, 2026. [LINK]
3. A. Mazumder, C. Garg, **T. Baruah**, P. Rathore, “*DIME: Deterministic Information Maximizing Autoencoder.*” Accepted at DeLTa @ International Conference on Learning Representations (ICLR), Singapore, 2025. [LINK]
4. A. Mazumder, **T. Baruah**, B. Kumar, R. Sharma, V. Pattanaik, P. Rathore, “*Learning Low-Rank Latent Spaces with Simple Deterministic Autoencoder: Theoretical and Empirical Insights.*” Accepted at IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), Hawaii, 2024. [LINK]
5. A. Mazumder, **T. Baruah**, A. Singh, P. Krishna, V. Pattanaik, P. Rathore, “*DeepVAT: A Self-Supervised Technique for Cluster Assessment for Image Datasets.*” Accepted at ViPrior @ IEEE/CVF International Conference on Computer Vision (ICCV), Paris, 2023. [LINK]
6. H. Wasnik, R. Gandhi, N. Patil, R. Behera, A. Golait, T. Patel, & **T. Baruah**, “*A Comprehensive Review of Molecular Biology and Genetics of Cataract.*” International Journal for Research in Applied Sciences and Biotechnology, 8(2), 11–20. [LINK]

Key Projects

Internship Project

Supervisor: Dr. Punit Rathore

CPS, IISc Bangalore (Project Page)

Jun 2023 – Jul 2023

- Formulated a novel autoencoder architecture that gives low-rank latent space.
- Used the architecture for downstream tasks such as classification, clustering, and generative tasks, outperforming baseline and SOTA autoencoders, including generative autoencoders such as VAE and flow-based autoencoders.

Master of Science Thesis Project

Supervisor: Dr. Prathosh AP

ECE, IISc Bangalore

Aug 2022 – Apr 2023

- **Representation Learning for Electronic Healthcare Records.**
- Developed augmentation techniques for Electronic Healthcare Records.
- Learned time-series representations using self-supervised modeling.
- Handled large-scale real-world electronic health record datasets, including MIMIC III & PhysioNet 2012.

Bachelor of Science Final Project

Supervisor: Dr. Haroon R Lone

EECS, IISER Bhopal

Dec 2021 – Apr 2022

• Calibrating Attention of Students in Classroom.

- Studied existing techniques and sensors, including galvanic, electroencephalography, and visual sensors.
- Deployed Raspberry Pi module and camera to collect student data from the classroom.
- Trained a real-time attentiveness detection model using collected data and computer vision techniques.

Summer Project: Water Rediscover

Project Instructor: Dr. Achintya N. Bezbaruah

North Dakota State University

Apr – Jul 2013

- Developed wastewater management model for recycling wastewater into potable water.
- Utilized locally available techniques and plants with cleansing properties.

Experience

Dept. of Cyber Physical Systems

Teaching Assistant: DA2180, Probabilistic Machine Learning: Theory and Applications

Course Instructor: Dr. Punit Rathore

Jan – May 2025

IISc Bengaluru

- Conducted weekly programming tutorials for topics taught in class.
- Assisted the instructor in formulating and evaluating assignments and exams.

Dept. of Data Science and Engineering

Teaching Assistant: DSE317/617, Machine Learning

Course Instructor: Dr. Tanmay Basu

Aug – Dec 2022

IISER Bhopal

- Delivered lectures on supplementary topics of the course.
- Mentored and guided students from the 2020 and 2019 batches throughout the course.
- Assisted the instructor in formulating and evaluating assignments and exams.

Dept. of Electrical Engineering & Computer Science

Teaching Assistant: ECS102, Introduction to Programming

Apr – May 2022

IISER Bhopal

- Mentored and guided students from the 2021 batch throughout the course.
- Conducted lab sessions to assist students with assignments and course-related difficulties.

AI and Cyber Physical System for Agriculture @ iHub – AWaDH

Workshop, Instructor: Dr. Neeraj Goel

May – Jul 2021

IIT Ropar

- Collected agricultural data via sensors and deployed them for AI model training.
- Developed Android app modules for agricultural applications.
- Completed capstone project: “Real-time Pest Detection using MobileNet-SSD & Farm Monitoring.”

Technical Skills

- **Programming:** Python, PyTorch, Distributed Data Parallel, Linux
- **Hardware Tools:** ESP32, Arduino, Raspberry Pi, IoT sensors

Relevant Courses

- **Mathematics:** Stochastic Modelling & Applications, Linear Algebra, Probability & Statistics, Calculus
- **Computer Science:** Data Structure & Algorithm, Discrete Maths, Internet of Things
- **Data Science:** Pattern Recognition & Neural Networks, Theory and Applications of Bayesian Learning, AI for Medical Image Analysis, Digital Image Processing
- **Biology:** Bio-molecules, Cell Biology, Biostatistics, Bio-informatics

- **Electrical and Electronics:** Signals & Systems, Principles of Communication, Electronic Devices, Electromagnetic Theory, Smart Sensing Technologies

Positions of Responsibility

Coordinator: Karate Club, IISER Bhopal *2018–2021*

Prefect: Students Council, Army Public School, Tezpur *2015–2016*

Extra-curriculars

- **Gold Medal:** Athletics 4*100m relay, Inter IISER Sports Meet, 2022
- **Visharad in Tabla:** 5-years classical training of tabla instrument